Q11 Find the coefficient of $w^2x^2y^2z^2$ in the expansion of

(a) $(w + x + y + z + 1)^{10}$

Solution.
The coefficient of $w^2x^2y^2z^2$ in the given expression after expansion is

$$\binom{10}{2,2,2,2,2} = \frac{10!}{2!2!2!2!2!} = 113400.$$

$\square$

(b) $(2w - x + 3y + z - 2)^{12}$
Similarly, the coefficient in the given expression after expansion is

$$\begin{aligned}\binom{12}{2,2,2,2,(12-2-2-2-2)} \cdot 2^2 \cdot (-1)^2 \cdot 3^2 \cdot 1^2 \cdot (-2)^{(12-2-2-2-2)} &= \frac{12!}{2!2!2!2!4!} \cdot 2^2 \cdot (-1)^2 \cdot 3^2 \cdot 1^2 \cdot (-2)^4 \\ &= 718502400.\end{aligned}$$

$\square$